

Reply by S. Höcht¹, W. Hinkelbein¹, N.E. Bechrakis², M. Foerster², H. Kluge³,
J. Heufelder³, D. Cordini³, H. Homeyer³ to theLetter by M. Fitzek in: *Strahlenther Onkol* 2007;183:49 (No. 1) (DOI 10.1007/s00066-007-9222-x)

In his letter to the editor, M. Fitzek raises severe doubts on the results reported in our publication on proton therapy of uveal melanomas.

We have to assume that the underlying problem is well known by both authors. Both have sufficient experience in the field. The definition of a tumor recurrence after proton therapy is quite complicated due to various regression patterns. Initially, and this is in a way our tribute to the learning curve, but certainly not a matter of inappropriate quality assurance in performing therapy, every patient with a single increase of the documented tumor prominence was counted as recurrence (Negroni, p. 55, lines 19–22 [2]). In the light of today's knowledge and as already pointed out in the discussion of Negroni's thesis on pages 55 and 56, this is inadequate because of uncertainties inherent in repeated ultrasound measurements during follow up and due to the secondary phenomenon of radiation-induced tumor edema in the early postirradiation period.

In an analysis performed in July 2005, the dataset of 283 patients showed an increase in tumor size in 38.5% of all patients during follow up. Omitting changes of ≤ 1 mm leaves 10% of all patients with an increase in size. Still not all of them should be counted as recurrence, as most of them later on showed a regression. Postirradiation edema is the most probable reason to explain this. The definition of recurrence therefore has to be more stringent: outgrowth of the base of the tumor as documented in fundus photographs and continuous increase in prominence of serial ultrasound measurement being some of the suggested major criteria. Negroni discusses these problems in detail in his thesis and states on page 56 lines 1–3 [2] that when applying an increase of at least 1 mm in tumor prominence, only six patients would be classified as having a recurrent tumor. Thus, the initial overreporting of suspected recurrences and their reduction in numbers in presentations published later is hereby explained sufficiently.

The next point to address is our database. It is not the database of the Negroni thesis. As already stated [1], in preparation of the paper, all patient charts were jointly reanalyzed by three of us (N.E.B., D.C., S.H.). Every fundus photograph, ultrasound measurement and other relevant findings were subjected to stringent reassessment.

Several years later we cannot rule out the possibility, that some very few patients who were initially judged as recurrent on the basis of initially not evident reversible posttherapeutic tumor edema were re-classified, based on our longer experience and in the knowledge of their further outcome.

The proton-beam-eye treatment facility in Berlin is open for other partners as well. Not all patients, but the vast majority of about 90%, have been treated by the Charité. The other hospital that treated patients there is the University of Essen Medical School, where N. Bornfeld is the ophthalmologist. Depending on whether these patients are included or not, the number of patients treated and the number of recurrences are different. Our analysis was restricted to those patients whose treatment planning, therapy and surveillance were done in Berlin [1].

M. Fitzek has left Essen and joined the group at Tufts University recently. He formerly was the radiation oncologist of the Essen therapy group who did the proton treatments in Berlin and at that time he was in close contact with most of the authors of the paper.

We have to apologize for a mistake that is stated correctly. J. Heese formerly worked at the proton-beam facility of the Hahn-Meitner Institute in Berlin, and contributed to the start-up work, but had left the group when the publication was prepared. On request of his former colleagues he was added as coauthor but accidentally did not receive the paper prior to publication.

References

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Address for Correspondence

Dr. Stefan Höcht
Department of Radiooncology and Radiotherapy
Charité – Universitätsmedizin Berlin
Campus Benjamin Franklin
Hindenburgdamm 30
12200 Berlin
e-mail: stefan.hoecht@charite.de

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¹ Klinik für Radioonkologie und Strahlentherapie, Charité Campus Benjamin Franklin, Berlin,

² Augenklinik, Charité Campus Benjamin Franklin, Berlin,

³ Augentumorthérapie, Hahn-Meitner-Institut Berlin.